ModelIt!

Teacher's Guide

Why Is Gold So Different from Iron?

How Atomic Structure Determines the Properties of Every Material

Quick Reference

Grade Level	Grade 08L2
NGSS Standard	MS-PS1-1, MS-PS1-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-PS1-1, MS-PS1-3

Develop models to describe the atomic composition of simple molecules and extended structures. Gather and make sense of information to describe that synthetic materials come from natural resources and impact society.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model relating atomic number and electron configuration to macroscopic material properties like reactivity and conductivity.
Disciplinary Core Idea	**PS1.A Structure and Properties of Matter** — Each pure substance has characteristic physical and chemical properties that can be used to identify it. Substances are made from different types of atoms, which combine in various ways.
Crosscutting Concept	**Structure and Function** — Students connect the invisible structure of atoms (electron configuration) to the visible function and behavior of materials (conductivity, reactivity, material properties).

CAST Testing Alignment

MULTIPLE CHOICE:

Element X has 11 protons and 1 electron in its outermost shell. Element Y has 18 protons and a completely full outermost shell. Which element is more likely to react violently with water, and why?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain why gold (atomic number 79) resists corrosion while iron (atomic number 26) rusts easily. Include the role of electron configuration in your explanation.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The periodic table organizes all known elements by their atomic structure, revealing patterns in properties that scientists have used for over 150 years. Understanding why gold resists corrosion while iron rusts, or why copper conducts electricity better than wood, requires looking at the atomic level — specifically at how electrons are arranged.

Atomic Number and Identity

Every element is defined by its atomic number — the number of protons in its nucleus. Hydrogen has 1, carbon has 6, iron has 26, gold has 79. Change the proton count and you change the element entirely. The atomic number also determines how many electrons surround the nucleus in a neutral atom, which in turn determines all chemical behavior.

Electron Configuration and Reactivity

Electrons occupy energy levels (shells) around the nucleus. The outermost shell determines reactivity. Atoms 'want' full outer shells — those with nearly empty outer shells (alkali metals like sodium) easily lose electrons, while those with nearly full shells (halogens like chlorine) easily gain electrons. Both are highly reactive. Noble gases already have full shells and barely react at all.

Why Gold Resists Corrosion

Gold is special among metals. Its large nucleus (79 protons) creates such a strong pull that inner electrons move at a significant fraction of the speed of light, causing relativistic effects that contract the inner shells and alter the outer electron energies. This makes gold's outer electrons unusually stable and resistant to chemical reactions — which is why gold jewelry from ancient Egypt still gleams today.

Conductivity and Electron Mobility

Metals conduct electricity because their outer electrons are loosely held and can flow freely through the material like a 'sea of electrons.' Silver is the best conductor, followed by copper, then gold. The arrangement and binding energy of outer electrons determine how freely they move. Non-metals hold their electrons tightly, making them poor conductors (insulators).

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify atomic number, electron configuration, reactivity, conductivity, and material properties as system components.

E **Establish Relationships**
Students determine that atomic number sets identity, electron configuration drives reactivity and conductivity, and these atomic-level factors produce observable material properties.

V **Visualize & Model**
Students build the atomic property model showing how atomic number and electron configuration determine reactivity, conductivity, and material properties.

E **Evaluate Outcomes**
Students compare elements with different atomic numbers and electron configurations to verify that outer-shell electrons predict reactivity and conductivity patterns.

R **Revise & Extend**
Students add melting point, electronegativity, or density to model more complex property relationships.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"One of these has been treasured for 5,000 years. The other crumbles in the rain. Both are metals. Why?"*

Watch For

Show gold and iron side by side. Ask: What makes them so different?

Slide 2: Learning Objectives

3 min

Say: *"Today we go inside the atom to discover why elements behave so differently."*

Watch For

Pre-teach atomic number (identity), electron configuration (behavior), reactivity (chemical willingness).

Slide 3: Big Question

3 min

Say: *"Both are made of protons, neutrons, and electrons. Same ingredients, wildly different results. Why?"*

Watch For

Show atomic diagrams. Count the protons. Ask: Does more protons always mean more reactive?

Slide 4: LEVER Framework

2 min

Say: *"We'll model how atomic structure creates the material properties we observe."*

Watch For

Preview that electron arrangement is the key to everything.

Slide 5: Activity 1: Components

8 min

Say: *"Atomic number and electron configuration are our inputs. Reactivity, conductivity, and material properties are our outputs."*

Watch For

Sort components. Discuss why atomic number and electron config are external (they define the element).

Slide 6: Activity 2: Connections

8 min

Say: *"Electron configuration drives everything — reactivity, conductivity, and the properties we can see and touch."*

Watch For

Map relationships. Highlight that outer-shell electrons are the most important factor.

Slide 7: Activity 3: Simulation

10 min

Say: *"Compare gold, iron, sodium, and neon. Watch how electron configuration predicts their behavior."*

Watch For

Students predict properties before running. Compare reactive vs. stable elements.

Slide 8: Discoveries

5 min

Say: *"The periodic table isn't random — it's organized by electron configuration, and properties follow predictable patterns."*

Watch For

Identify groups: alkali metals (reactive), noble gases (stable), transition metals (variable).

Slide 9: STEM Challenge

12 min

Say: *"Engineers choose materials based on atomic properties. Your turn — recommend the best material for each job."*

Watch For

Teams analyze applications, select materials using model evidence, and present recommendations.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Atomic Number, Electron Configuration
- **INTERNAL:** Reactivity, Conductivity, Material Properties

Relationships

- **Atomic Number to Electron Configuration:** POSITIVE (+) — More protons means more electrons in a neutral atom, filling additional energy levels and creating more complex configurations.
- **Electron Configuration to Reactivity:** POSITIVE/NEGATIVE (+/-) — Elements with nearly empty or nearly full outer shells are highly reactive. Elements with full outer shells have near-zero reactivity. The relationship depends on how close the outer shell is to being full or empty.
- **Electron Configuration to Conductivity:** POSITIVE (+) — Elements with loosely held outer electrons (metals) allow electron flow, creating high conductivity. Tightly held electrons (non-metals) resist flow.
- **Reactivity to Material Properties:** POSITIVE (+) — High reactivity leads to observable properties like tarnishing, corrosion, or vigorous reactions with water and acids.
- **Conductivity to Material Properties:** POSITIVE (+) — High conductivity correlates with metallic luster, malleability, and the ability to transmit heat — all observable material properties.

Key Discoveries

- Atomic number determines the element's identity — change the proton count and you change the element entirely
- Electron configuration, especially the outermost shell, determines how an element bonds and reacts
- Elements with nearly full or nearly empty outer shells are the most reactive
- Elements with full outer shells (noble gases) are extremely stable and barely react
- Conductivity depends on how freely outer electrons can move — metals with loosely held electrons conduct well

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"All metals behave the same way"	Metals vary enormously in reactivity, conductivity, hardness, and other properties based on their atomic structure. Sodium explodes in water while gold sits in seawater for centuries unchanged. Mercury is a liquid at room temperature while tungsten doesn't melt until 3,422°C. Electron configuration explains these differences.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bigger atoms are always more reactive"	Reactivity depends on electron configuration, not size. Neon (10 protons) is completely unreactive because its outer shell is full, while sodium (11 protons, just one more) is explosively reactive because it has a single lonely outer electron. Size alone tells you nothing about reactivity.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Electrons orbit like planets around the sun"	Electrons don't travel in neat circular orbits. They exist in probability clouds (orbitals) — regions where they are likely to be found. The 'shell' model is a useful simplification for understanding energy levels and reactivity, but real electron behavior is described by quantum mechanics.